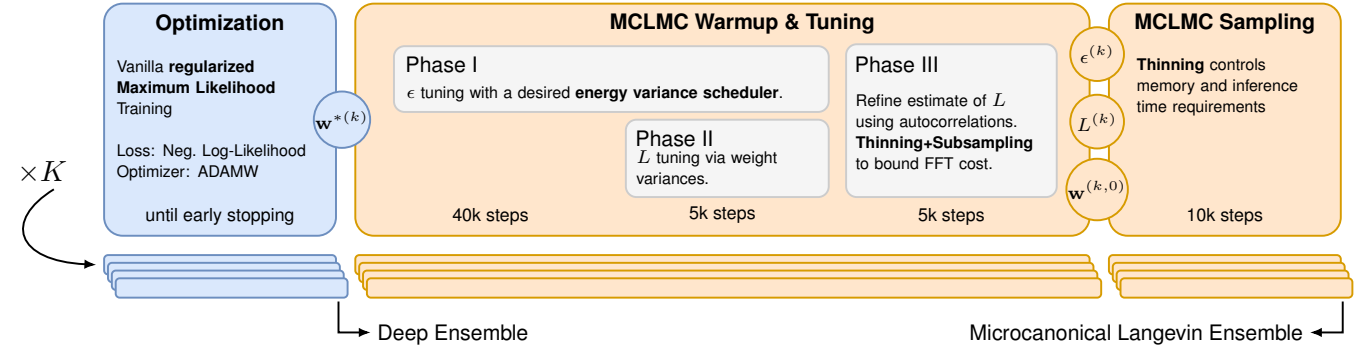

Optimization
Vanilla **regularized Maximum Likelihood** Training
Loss: Neg. Log-Likelihood
Optimizer: ADAMW
until early stopping
$\mathbf{w}^{*(k)}$
MCLMC Warmup & Tuning
Phase I
ϵ tuning with a desired **energy variance scheduler**.
Phase II
L tuning via weight variances.
Phase III
Refine estimate of L using autocorrelations.
Thinning+Subsampling to bound FFT cost.
40k steps
5k steps
5k steps
$\epsilon^{(k)}$
$L^{(k)}$
$\mathbf{w}^{(k,0)}$
MCLMC Sampling
Thinning controls memory and inference time requirements
10k steps
$\times K$
Deep Ensemble
Microcanonical Langevin Ensemble